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EFFECTS OF OXYTOCIN ON CARDIAC FUNCTION IN LIPOPOLYSACCHARIDE INDUCED SEPSIS

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USE OF STROKE VOLUME VARIATION TO GUIDE FLUID THERAPY IN SEPTIC SHOCK FOR PREVENTION OF ACUTE KIDNEY INJURY

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Parameter of interest	Conventional therapy ($n = 48$)	Stroke volume variation ($n = 53$)	P value
Age	51 $\pm$ 11.3	52.5 $\pm$ 11.7	0.6
Sex (males, %)	36 (75%)	38 (71%)	0.7
APACHE II score	20.1 $\pm$ 6	21 $\pm$ 7.1	0.6
Predicted death rate	37 $\pm$ 18.9	40.3 $\pm$ 22	0.44
Incidence of AKI	29 (60.4%)	21 (39.6%)	0.037
Need for RRT	21 (43.8%)	15 (28.3%)	0.1
Length of ICU stay	7.2 $\pm$ 3.1	6.8 $\pm$ 3.4	0.83
ICU mortality	19 (39.6%)	18 (34%)	0.56

CONCLUSIONS. Fluid management in septic patients according to SVV significantly reduces the incidence of AKI. Although, a trend towards reduction of RRT requirement, length of stay in ICU and overall mortality was seen, it did not translate into statistically significant reduction, in our study. Multi-center study with a larger cohort may be required to show such benefit.**REFERENCE(S).** 1. Reuter DA et al (2003) Usefulness of left ventricular stroke volume variation to assess fluid responsiveness in patients with reduced cardiac function. Crit Care Med 31:1300–1404.**GRANTS.** Nil.

0105

SYMPATHETIC TONE AND VASCULAR K_{ATP} CHANNEL ACTIVITYY.-L. Chan^{1,2,3}, V. Taylor², R. Stidwill², L. H. Clapp¹, M. Singer^{1,2}¹Department of Medicine, University College London, London, UK, ²Wolfson Institute for Biomedical Research, University College London, London, UK, ³Department of Emergency Medicine, Chang Gung Memorial Hospital/Chang Gung University, Taoyuan, Taiwan, Republic of China**INTRODUCTION.** Vascular K_{ATP} channel hyperactivity is implicated in the cardiovascular collapse of septic shock, although glibenclamide, a classical channel inhibitor acting through the sulphonylurea receptor subunit, did not affect norepinephrine (NE) responses in septic shock patients [1]. Moreover, PNU-37883A (PNU), an inhibitor acting through the pore-forming subunit of the channel, did not affect BP in our awake peritonitis rat model [2]. Given that vasoconstrictors, including NE, inhibit K_{ATP} channel activity [3], we speculate that the high sympathetic tone seen in sepsis [4] may counteract K_{ATP} channel opening.**OBJECTIVE.** To investigate the effect of sympathetic modulation on BP changes induced by vascular K_{ATP} channel pore blockade during sepsis.**METHODS.** Fluid-resuscitated male Wistar rats (~300 g) were subjected to faecal peritonitis for 6 h. Changes in BP induced by NE [0.5 μ g/(kg min) i.v.], $\pm$ PNU (1.5 mg/kg i.v.), were compared in awake animals pretreated with the autonomic ganglion blocker pentolinium [2 mg/kg bolus, 2 mg/(kg h) i.v.], or sodium nitroprusside [SNP; 50 μ g/(kg min) i.v.], a nitric oxide (NO) donor and vasodilator, of which the dose was chosen to achieve similar BP fall compared to pentolinium.**RESULTS.** Peritonitis rats showed pressor hyporeactivity to NE (Table 1). Pentolinium induced a larger drop in BP in peritonitis rats compared to controls, implying raised sympathetic tone. PNU elevated BP more in pentolinium-pretreated septic animals. SNP, however, decreased PNU's pressor effect. This may be related to the exogenous NO or vasodilation from SNP, or the induced increase in sympathetic tone.

TABLE 1 BP of sham and 6-h faecal peritonitis rats treated with drugs

BP (mmHg)	Sham ($n = 8$)	Peritonitis ($n = 8$)	BP (mmHg)	Sham ($n = 4$)	Peritonitis ($n = 4$)
Baseline	118.6 $\pm$ 4.7	134.6 $\pm$ 3.2	Baseline	122.9 $\pm$ 4.0	133.3 $\pm$ 5.8
NE	163.2 $\pm$ 4.9**	141.0 $\pm$ 3.7	SNP	83.9 $\pm$ 3.3**	101.3 $\pm$ 8.1**
Pentolinium	78.1 $\pm$ 4.1**	74.7 $\pm$ 5.2**	+PNU	91.1 $\pm$ 4.2**	110.4 $\pm$ 6.1**
+NE	162.2 $\pm$ 4.7** ^{††}	138.1 $\pm$ 6.3** ^{††}	+PNU + NE	116.3 $\pm$ 7.6	116.6 $\pm$ 4.9**
+PNU	120.1 $\pm$ 4.4 ^{††}	101.8 $\pm$ 3.2** ^{††}			
+PNU + NE	164.6 $\pm$ 6.1** ^{††}	136.3 $\pm$ 4.0** ^{††}			

Data were shown as mean $\pm$ SEM** $p < 0.01$ versus baseline, [†] $p < 0.05$, ^{††} $p < 0.01$ versus sham, ^{††} $p < 0.01$ versus pentolinium**CONCLUSIONS.** High sympathetic tone appears to mask the pressor effect of vascular K_{ATP} channel inhibition in sepsis. This may explain lack of efficacy of channel inhibitors in clinical studies.**REFERENCE(S).** 1. Warrillow S et al (2004) Crit Care Med 34:980–985. 2. Chan YL et al (2008) Intensive Care Med 34:S121. 3. Bonev AD, Nelson MT (1996) J Gen Physiol 108:315–323. 4. Leinhardt DJ et al (1993) Am J Physiol 265:E284–E288.**GRANT ACKNOWLEDGEMENT.** Chang Gung Medical Research Project CMRPG270091.

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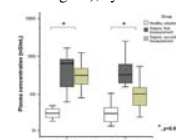
SHEDDING OF VASCULAR ENDOTHELIAL GLYCOCALYX LAYER AND SUBLINGUAL MICROCIRCULATION IN SEVERE SEPSIS

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Variables during video microscopy

	Temp (°C)	Hct (%)	MAP (mmHg)	HR (min)	SvO ₂ (%)	CVP (mmHg)	PCWP (mmHg)	Norepinephrine [μ g/(kg min)]	MFI small
Sepsis	36.7	31	73	101	74	15	13	0.35	2.9
day 1	(36.4–37.1)	(29–36)	(70–82)	(83–110)	(68–78)	(13–15)	(12–15)	(0.16–0.63)	(2.8–3)
Sepsis	36.3	27	89	74	69	11	14	0.02	2.9
day 4	(36.0–36.7)	(26–31)	(78–94)	(71–86)	(65–78)	(10–14)	(12–16)	(0–0.06)	(2.8–3)
Healthy	36.8	41	95	60					2.8
volunteers	(36.5–37.1)	(39–42)	(89–100)	(55–65)					(2.7–3)

Plasma levels of endothelial glycocalyx markers were significantly elevated in severe sepsis at day 1 [syndecan-1: 642.5 ng/ml (IQR 159.3–825.6 ng/ml), hyaluronan: 319.5 ng/ml (IQR 199.9–611.5 ng/ml)] and at day 4 [syndecan-1: 307.9 ng/ml (IQR 187.7–474.2 ng/ml), hyaluronan: 102.1 ng/ml (IQR 50.1–152.2 ng/ml)] as compared to values measured from healthy volunteers [syndecan-1: 30.5 ng/ml (IQR 23.7–41.1 ng/ml), hyaluronan: 30.0 ng/ml (IQR 17.9–43.5 ng/ml)].



Plasma markers of vascular endothelial glycocalyx

CONCLUSION. Shedding of vascular endothelial glycocalyx layer was detected by plasma samples of syndecan-1 and hyaluronan, but simultaneous derangement of microcirculation could not be detected by MFI. A more detailed study of microcirculatory indices will be carried out.